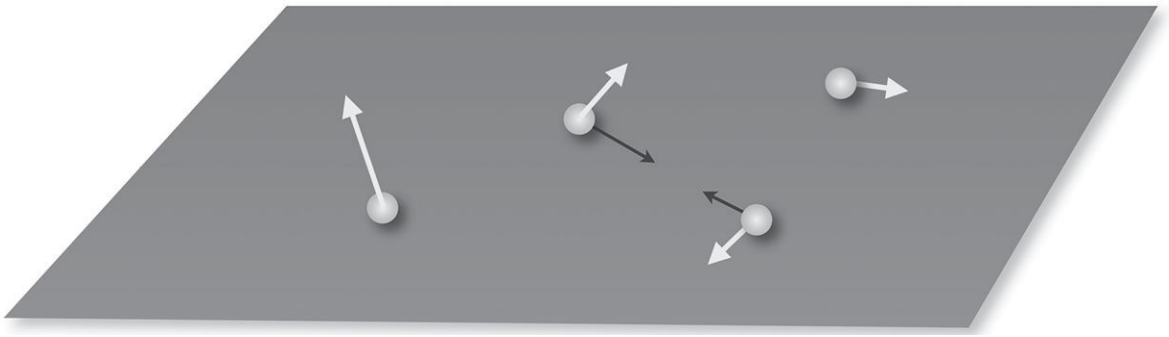


Figure 2.2 The world of Newton: particles which move in space, as time passes, attracted by forces.

Newton acknowledges without hesitation the debt that the new science owes to the science of antiquity. In the first lines of his *The System of the World*, for example, he attributes to antiquity (correctly) the origins of the idea at the basis of the Copernican revolution: 'It was the opinion of the ancient philosophers that in the highest parts of the world the stars remain fixed and motionless, and that the Earth turns around the Sun'; although he is a bit confused as to who did what in the past, and quotes – sometimes appositely, sometimes out of context – Philolaus, Aristarchus of Samos, Anaximander, Plato, Anaxagoras, Democritus and (!) 'the learned Numa Pompilius, king of the Romans'.

The power of the new Newtonian intellectual frame-work proves to be beyond all expectation. The entire technology of the nineteenth century and of our own modern world rests largely upon Newton's formulae. Three centuries have passed, but it's still thanks to theories based upon Newton's equations that today we build bridges, trains and skyscrapers, engines and hydraulic systems; that we know how to fly planes, make weather forecasts, predict the existence of a planet before seeing it and send spaceships to Mars … The modern world would not have been born without passing by way of Newton's little moon.

A new conception of the world, a way of thinking which raised the enthusiasm of the Enlightenment of Voltaire and Kant, an effective way for predicting the future: this has been and continues to be the immense legacy of the Newtonian revolution.

It seemed thus that the final key to understanding reality had been discovered: the world consists only of a great, infinite space where, as time passes, particles move and attract one another by means of forces. We can write precise equations that describe these forces. These equations prove to be immensely effective. Still in the nineteenth century, people said that Newton was not only one of the most intelligent and far-seeing of men but also the most fortunate – because there is only one system of fundamental laws, and he was given the good fortune of discovering it. Everything seemed clear.